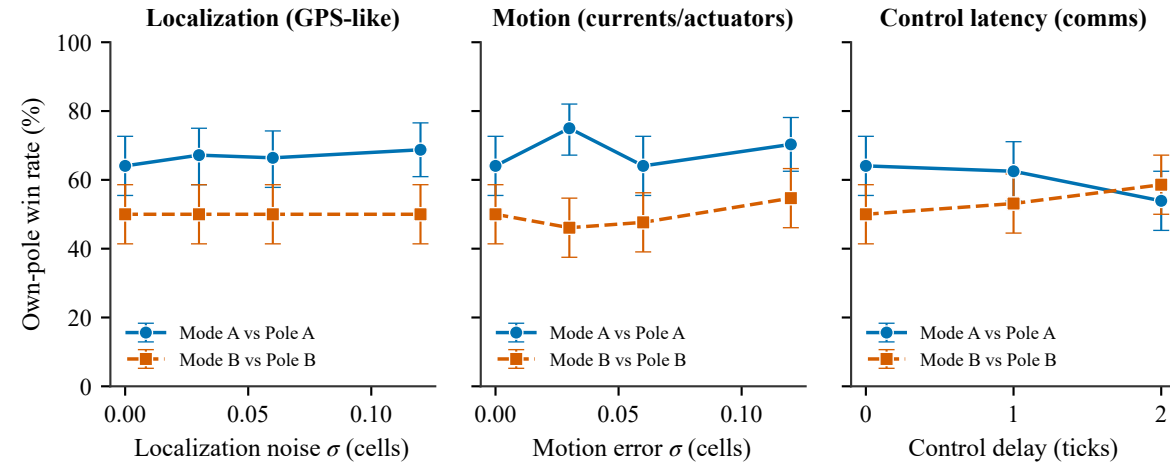




*Sim-to-real diagnostic dose-response ( $n=128$ ). Families map to physical deployment: localization  $\rightarrow$  GPS/pose uncertainty; motion  $\rightarrow$  currents/wind/actuator mismatch; delay  $\rightarrow$  communications/control latency. Matched numeric schedule for localization/motion (0.03/0.06/0.12 cells); delay 0/1/2 ticks. Error bars: 95% bootstrap CIs.*